

# Statistics for Land Management

---

## Science

May, 2026

# Statistics for Land Management (A07550)

**Short Title:** Statistics for Land Management  
**Faculty** Science and Computing  
**Department** Science  
**Cluster** Mathematics and Physics  
**Author** AHENNESSY  
**Credits** 5

**Level:** Advanced

## Description of Module / Aims

The module is designed to introduce key concepts in statistics for research in agriculture, forestry and horticulture. Students will learn how statistical tools can be applied by modelling relevant data on statistical software, as well as analysing and interpreting the results.

## Programmes

|           |                                                              |           |
|-----------|--------------------------------------------------------------|-----------|
| SCIE-0069 | BSc (Hons) in Land Management (in Agriculture) (WD_SBSAG_B)  | 1 / 1 / M |
| SCIE-0069 | BSc (Hons) in Land Management (in Forestry) (WD_SBSFO_B)     | 1 / 1 / M |
| SCIE-0069 | BSc (Hons) in Land Management (in Horticulture) (WD_SBSHO_B) | 1 / 1 / M |

## Indicative Content

- Basic statistics: data collection methods; presentation of data; measures of center and spread
- Regression: least squares line; Pearson's correlation coefficient; the coefficient of determination; rank correlation
- Inferential statistics: sampling distributions; hypothesis testing; confidence intervals; ANOVA; Chi-squared tests

## Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Choose appropriate data collection methods and apply relevant graphical and analytical techniques for the investigation of data.
2. Formulate regression and correlation techniques in appropriate sets of data.
3. Set up a broad range of analytical statistical techniques and interpret the results correctly.
4. Select appropriate statistical software tools to analyse and interpret data.

## Learning and Teaching Methods

- Lectures.
- Practical labs.

## Assessment Methods

|                           | Weighing | Outcomes Assessed |
|---------------------------|----------|-------------------|
| Final Written Examination | 75%      | 1,2,3             |
| Continuous Assessment     | 25%      |                   |
| Assignment                | 25%      | 1,2,3,4           |

## Assessment Criteria

- <40%:** Very limited knowledge and understanding of statistics.
- 40%-49%:** Demonstrate a limited knowledge of statistics.
- 50%-59%:** Demonstrate satisfactory general knowledge of statistics.
- 60%-69%:** Demonstrate sound knowledge of statistics. Show ability to analyse and logically argue in an effective and mature style.
- 70%-100%:** Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of statistics.

## Learning Modes

| Learning Mode        | F/T Hours | P/T Hours |
|----------------------|-----------|-----------|
| Lecture              | 36        |           |
| Practical            | 12        |           |
| Independent Learning | 87        |           |

## Essential Material

- "Statistics for Land Management moodle page." <https://moodle.wit.ie>

## Supplementary Material

- Nageswara, R. \_Statistics for Agricultural Sciences\_. India: BS Publications, 2007.
- Samuels, M. \_Statistics for the life sciences\_. 4th ed. New Jersey: Pearson Education, 2012.

## Requested Resources

- Room Type: Computer Lab